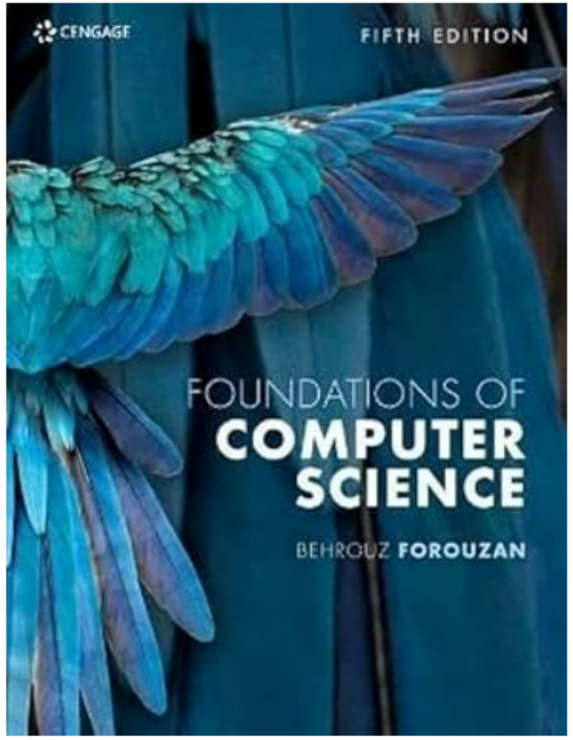
2026 Spring

Course ID: 1142-1764

Foundations of Computer Science

09-Networking and Content Delivery

The Textbook



Foundations of Computer Science Paperback

作者 Behrouz Forouzan (Author)

The symposium was held Oct.-Nov. 1989, Research Triangle Park, North Carolina. One hundred papers in theoretical computer science treat dispersers, deterministic amplification, and weak random sources; efficient NC algorithms for set cover with applications to learning and geometry; the inverse of automorphism in polynomial time; and speeding-up linear programming using fast matrix multiplication. Acidic paper; no subject index. Annotation copyright Book News, Inc. Portland, Or.

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語言	尺寸	ISBN-10	ISBN-13
English	18.9 x 24.7 x 2.6 cm	1473787327	978-1473787322

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AWS ACADEMY · CLOUD FOUNDATIONS

Flip Classroom

Padlet Collaborative Learning

Introduction to Computer Science | Modules 5 – 10

- M5 Networking
- M6 Compute
- M7 Storage
- M8 Databases
- M9 Architecture
- M10 Auto Scaling

📌 *Your Group Builds the Knowledge Base — Together*

🔄 What Is a Flip Classroom?

1 BEFORE Class

Study the module
watch videos,
read materials at home

→

2 PREPARE Padlet

Your group fills in
all 5 sections of
the Padlet template

→

3 IN Class

Present & discuss
answer classmates'
questions (Q&A)

→

4 REFLECT

Respond to peer
comments, update
the Padlet

💡 Padlet is the shared workspace where your group documents your learning for each module — visible to the whole class!

Your Padlet Template — 5 Sections

S1

Learning Objectives

State the key learning goals of the module.
What should you understand by the end?

S2

Core Concepts

Summarize essential technical concepts, terminology, and architectures.

S3

Labs

Document hands-on AWS lab steps, screenshots, and observations.

S4

Challenges

Record difficult questions, errors you encountered, and how you solved them.

S5

Q&A & Peer Support

Post questions for the class.
Answer and support your peers.

Tips for a Great Padlet

Add images & diagrams, not just text

Link to AWS documentation & tutorials

Use clear headings in each column

Respond to at least 2 peers in S5

M5 Networking & Content Delivery

📘 Core Concepts to Explore

- VPC, Subnets & CIDR Blocks
- Internet Gateway & Route Tables
- Security Groups vs. NACLs
- Amazon CloudFront CDN
- Elastic Load Balancing (ELB)

📌 Your Padlet Tasks

S1	**Learning Objectives**	List 3–5 objectives for this module
S2	**Core Concepts**	Diagram a VPC with public & private subnets
S3	**Lab Summary**	AWS Lab – Build a VPC; screenshot your architecture
S4	**Challenges**	Why use private subnets? What confused you about NACLs?
S5	**Q&A**	Post 1 question, reply to 2 peers

M6 Compute

📘 Core Concepts to Explore

- EC2 Instance Types & AMIs
- Pricing Models (On-Demand, Reserved, Spot)
- Elastic Beanstalk & Lambda
- Auto Scaling Groups Intro
- AWS Pricing Calculator

📌 Your Padlet Tasks

S1	**Learning Objectives**	List 3–5 objectives for this module
S2	**Core Concepts**	Compare EC2 pricing models in a table
S3	**Lab Summary**	Launch & connect to an EC2 instance; log steps
S4	**Challenges**	What's the trade-off between Reserved vs. Spot instances?
S5	**Q&A**	Post 1 question, reply to 2 peers

M7 Storage

📘 Core Concepts to Explore

- S3 Buckets, Objects & Classes
- EBS vs. EFS vs. S3
- S3 Lifecycle Policies
- Glacier for Archiving
- Storage Security & Encryption

📌 Your Padlet Tasks

S1	**Learning Objectives**	List 3–5 objectives for this module
S2	**Core Concepts**	Decision tree: when to use S3, EBS, or EFS?
S3	**Lab Summary**	Create S3 bucket, upload files, set permissions
S4	**Challenges**	When would you choose Glacier over S3 Standard?
S5	**Q&A**	Post 1 question, reply to 2 peers

M8 Databases

📘 Core Concepts to Explore

- RDS & Multi-AZ Deployments
- Aurora vs. MySQL/PostgreSQL
- DynamoDB (NoSQL)
- Elasticache for Caching
- Database Migration Service

📌 Your Padlet Tasks

S1	**Learning Objectives**	List 3–5 objectives for this module
S2	**Core Concepts**	SQL vs. NoSQL comparison card
S3	**Lab Summary**	Create RDS instance; document endpoint & SG config
S4	**Challenges**	When is DynamoDB preferred over RDS? Give an example.
S5	**Q&A**	Post 1 question, reply to 2 peers

M9 Cloud Architecture

📘 Core Concepts to Explore

- AWS Well-Architected Framework (6 Pillars)
- High Availability & Fault Tolerance
- Disaster Recovery Strategies
- Loose Coupling & Microservices
- Cost Optimization Principles

📌 Your Padlet Tasks

S1	**Learning Objectives**	List 3–5 objectives for this module
S2	**Core Concepts**	Map the 6 pillars with one AWS service each
S3	**Lab Summary**	Design a HA architecture diagram using AWS icons
S4	**Challenges**	How does loose coupling improve reliability?
S5	**Q&A**	Post 1 question, reply to 2 peers

M10 Auto Scaling & Monitoring

📘 Core Concepts to Explore

- EC2 Auto Scaling Policies
- CloudWatch Metrics & Alarms
- CloudTrail Audit Logs
- Trusted Advisor Checks
- SNS Notifications

📌 Your Padlet Tasks

S1	**Learning Objectives**	List 3–5 objectives for this module
S2	**Core Concepts**	Draw an Auto Scaling trigger flow diagram
S3	**Lab Summary**	Set up CloudWatch alarm; test with CPU load
S4	**Challenges**	What metric would you monitor for a web app under load?
S5	**Q&A**	Post 1 question, reply to 2 peers

Evaluation Criteria

30%

Content Quality

Accuracy, depth, and clarity of each of the 5 Padlet sections

20%

Collaboration

Evidence that all group members contributed content and ideas

20%

Presentation

Clear in-class explanation, time management, visual aids used

15%

Peer Interaction (S5)

Quality of questions posted and responses given to classmates

15%

Creativity & Format

Use of images, diagrams, links, and original visual design

Total: 100% · Padlet link submitted counts as evidence for continuous assessment

Ready to Start?

Your group's Padlet is your team's public knowledge contribution to the entire Introduction to Computer Science class.

Step 1

Access the Padlet template link on LMS

Step 2

Form your group & claim your module

Step 3

Start filling in all 5 sections now!

Good luck, and have fun exploring AWS Cloud!

Grading

The report's grading includes:

- 20% teacher's score
- 20% group leader's score
- 60% peer evaluation.

Group leaders:
Module 6 - 411170341 林家畯
Module 7 - 411416878 鄭鎧文
Module 8 - 411416886 蔡美彥
Module 9 - 411416991 許原盛
Module 10 - E11420600 哈里哈蘭